

Linear Regression Multiple Variables

```
In [18]: import pandas as pd
from sklearn.linear_model import LinearRegression

# 1. Load dataset
df = pd.read_csv('homeprices.csv')
df
```

```
Out[18]:
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	NaN	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4100	6.0	8	810000

--- HANDLE MISSING VALUES ---

```
In [33]: # Option: Fill missing values with column means
#df = df.fillna(df.median())
df.bedrooms = df.bedrooms.fillna(df.bedrooms.median())
df
```

```
Out[33]:
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	4.0	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000
5	4100	6.0	8	810000

Define Independent Variables (Features) and Dependent Variable (Target)

```
In [20]: # Independent Variables (Features)
X = df[['area', 'bedrooms', 'age']]
print("Independent Variables (Features) :")
```

```
print(X)

# Dependent Variable (Target)
y = df['price']
print("Dependent Variables (Target) :")
print(y)
```

Independent Variables (Features) :

	area	bedrooms	age
0	2600	3.0	20
1	3000	4.0	15
2	3200	4.0	18
3	3600	3.0	30
4	4000	5.0	8
5	4100	6.0	8

Dependent Variables (Target) :

0	550000
1	565000
2	610000
3	595000
4	760000
5	810000

Name: price, dtype: int64

Initialize and train the model

```
In [43]: model = LinearRegression()
          model.fit(X, y)           #fit means training
```

```
Out[43]: ▼ LinearRegression ⓘ ?
```

► Parameters

► Fitted attributes

Model coeff. & Intercept

```
In [44]: model.coef_
```

```
Out[44]: array([ 112.06244194, 23388.88007794, -3231.71790863])
```

```
In [45]: model.intercept_
```

```
Out[45]: np.float64(221323.00186540402)
```

Prediction

CASE 1 : Find price of home with 3000 sqr ft area, 3 bedrooms, 40 year old

In [46]: `model.predict([[3000,3,40]])`

E:\anaconda\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

Out[46]: `array([498408.25158031])`

112.06 x + 23388.88 y -3231.71 z + 221323 = Price (498408.2)

In [47]: `112.06244194*3000 + 23388.88007794*3 + -3231.71790863*40 + 221323.00186540384`

Out[47]: `498408.25157402386`

CASE 2 : Find price of home with 2500 sqr ft area, 4 bedrooms, 5 year old

In [48]: `model.predict([[2500, 4, 5]])`

E:\anaconda\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

Out[48]: `array([578876.03748933])`

Final

In [49]: `# 6. Make a prediction for a new home
new_homes = pd.DataFrame([[2500, 4, 5]], columns=['area', 'bedrooms', 'age'])

Predict, round to 0 decimals, and convert to integer
new_homes['predicted_price'] = model.predict(new_homes).round(0).astype(int)

Print
print("--- Prediction Result ---")
print(new_homes)`

```
--- Prediction Result ---
   area  bedrooms  age  predicted_price
0  2500         4    5           578876
```

In []: